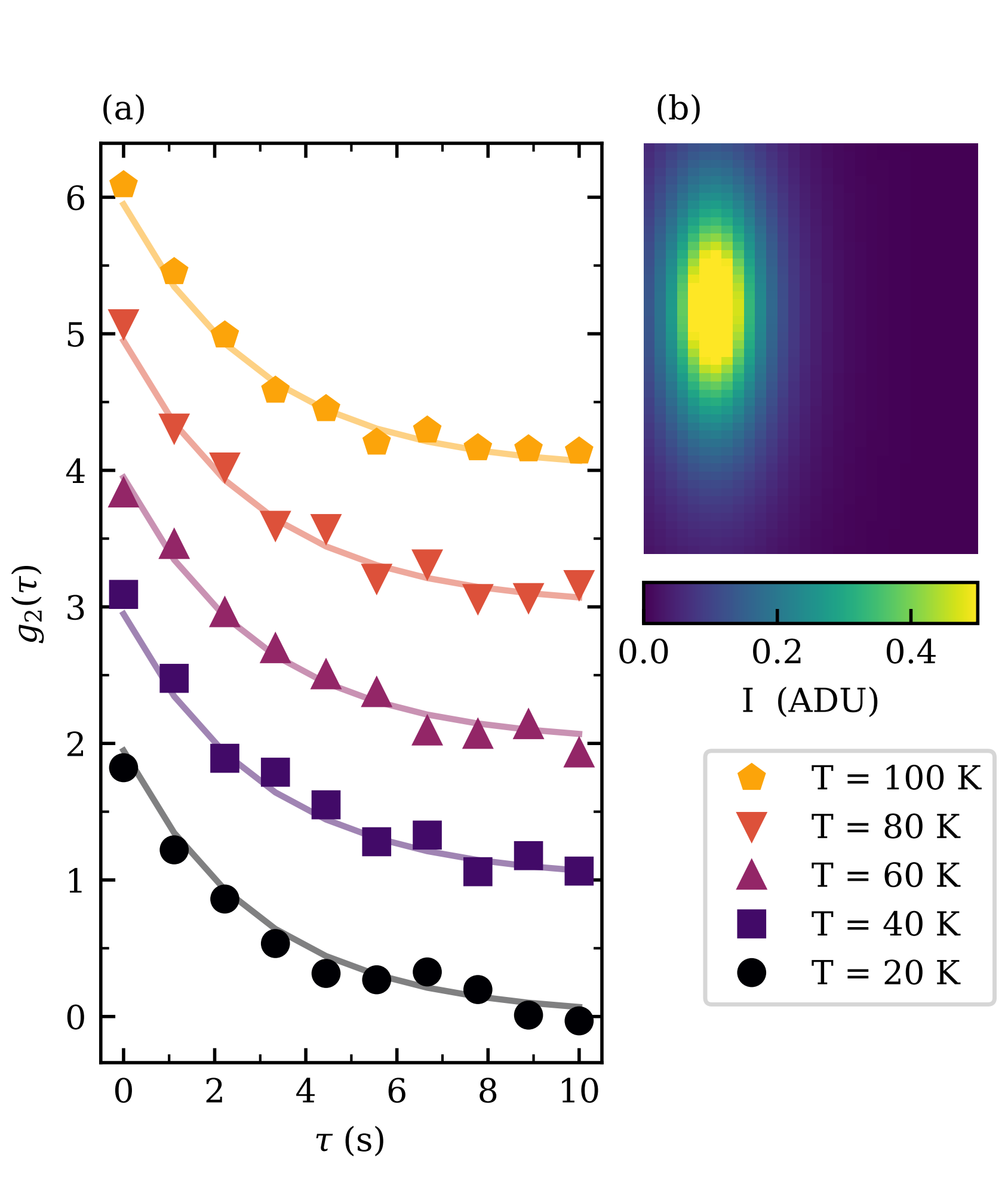
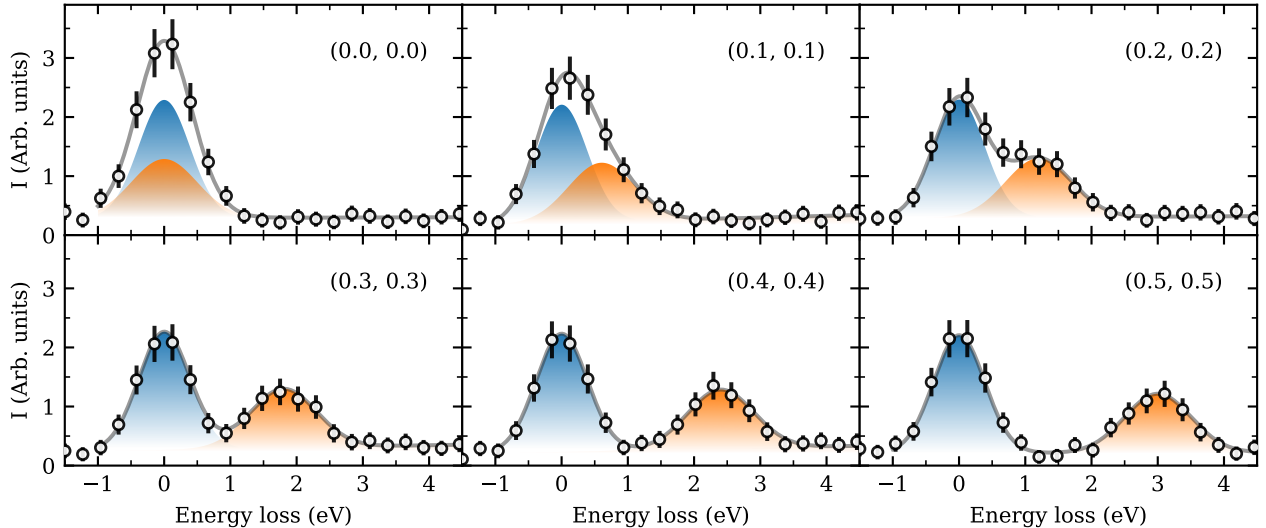
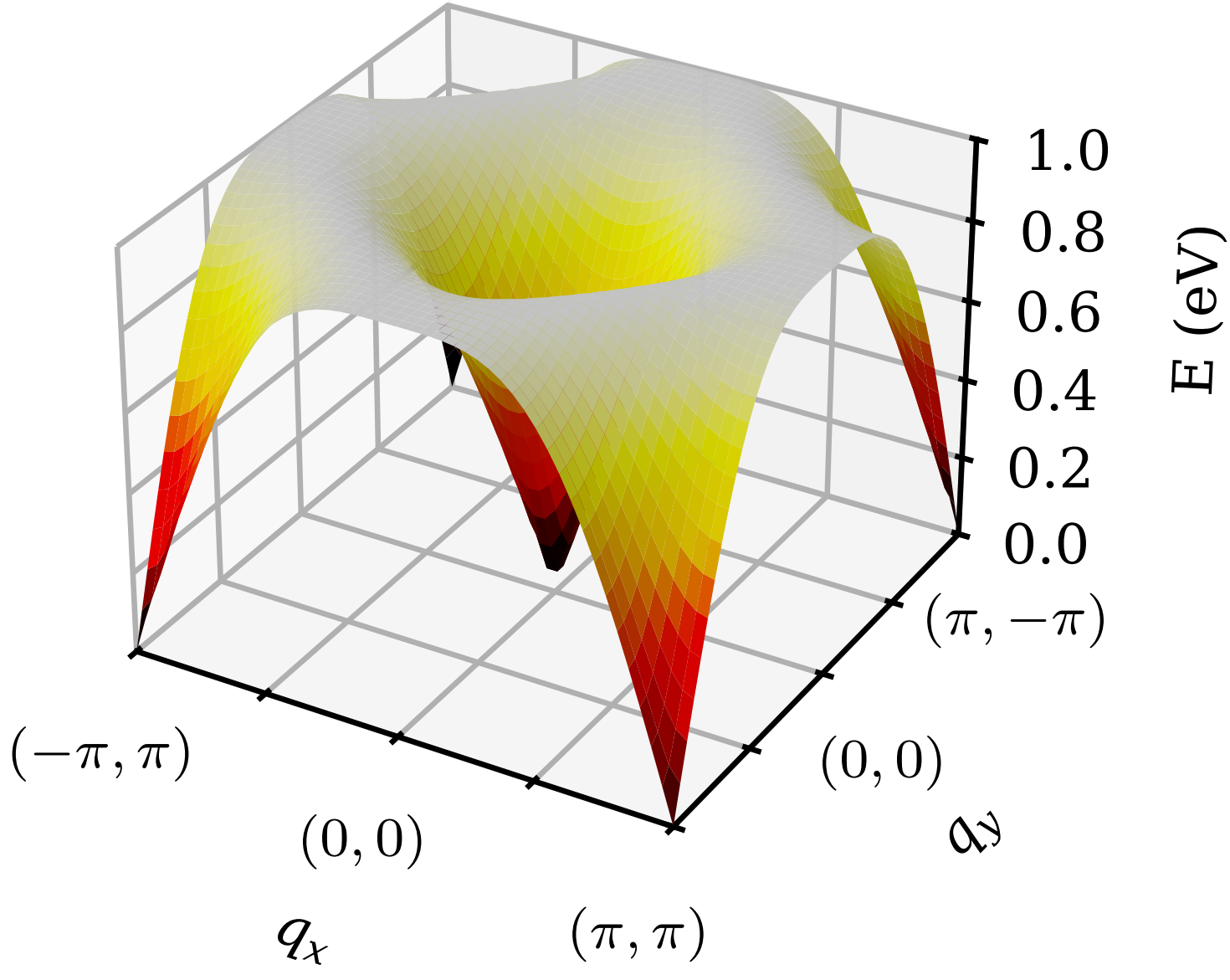
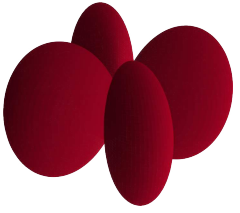

(a)
(b)
$g_2(\tau)$
τ (s)
0
1
2
3
4
5
6
0
2
4
6
8
10
0.0
0.2
0.4
I (ADU)
T = 100 K
T = 80 K
T = 60 K
T = 40 K
T = 20 K


(0.0, 0.0)
(0.1, 0.1)
(0.2, 0.2)
(0.3, 0.3)
(0.4, 0.4)
(0.5, 0.5)
I (Arb. units)
I (Arb. units)
0
1
2
3
−1
0
1
2
3
4
Energy loss (eV)
Energy loss (eV)
Energy loss (eV)


1.0
0.8
0.6
0.4
0.2
0.0
E (eV)
$(\pi, -\pi)$
$(0, 0)$
q_y
$(-\pi, \pi)$
$(0, 0)$
q_x
(π, π)

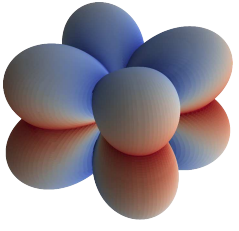
$L_z = 0$



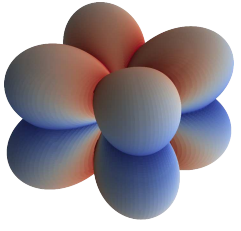
$L_z = 1$

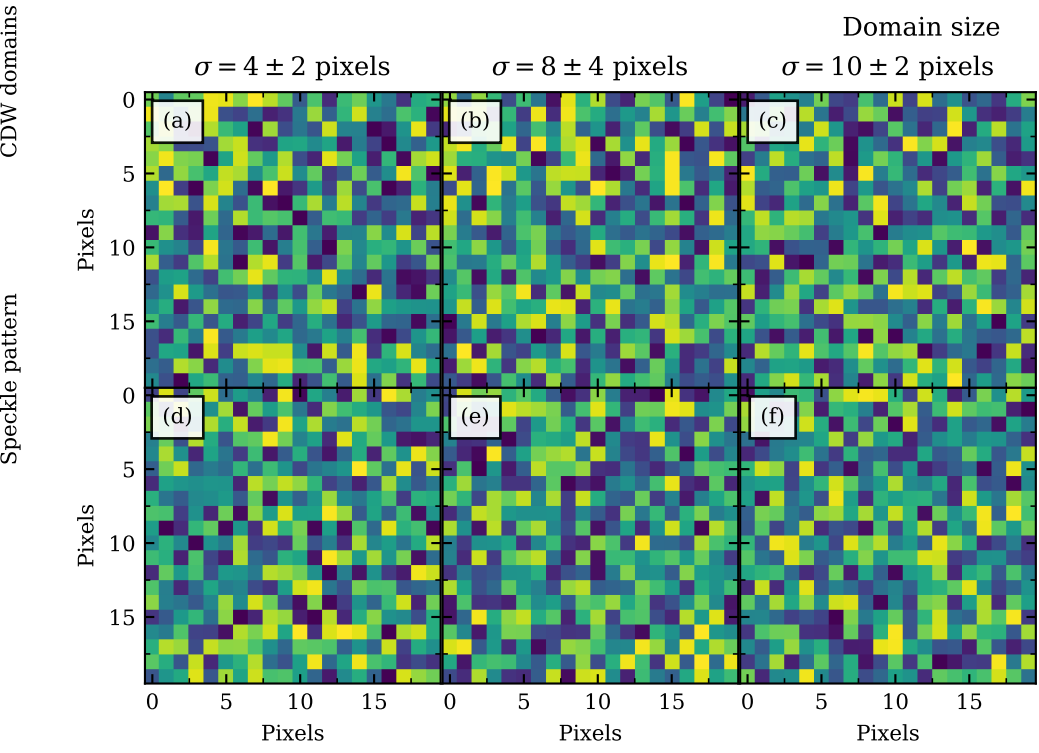


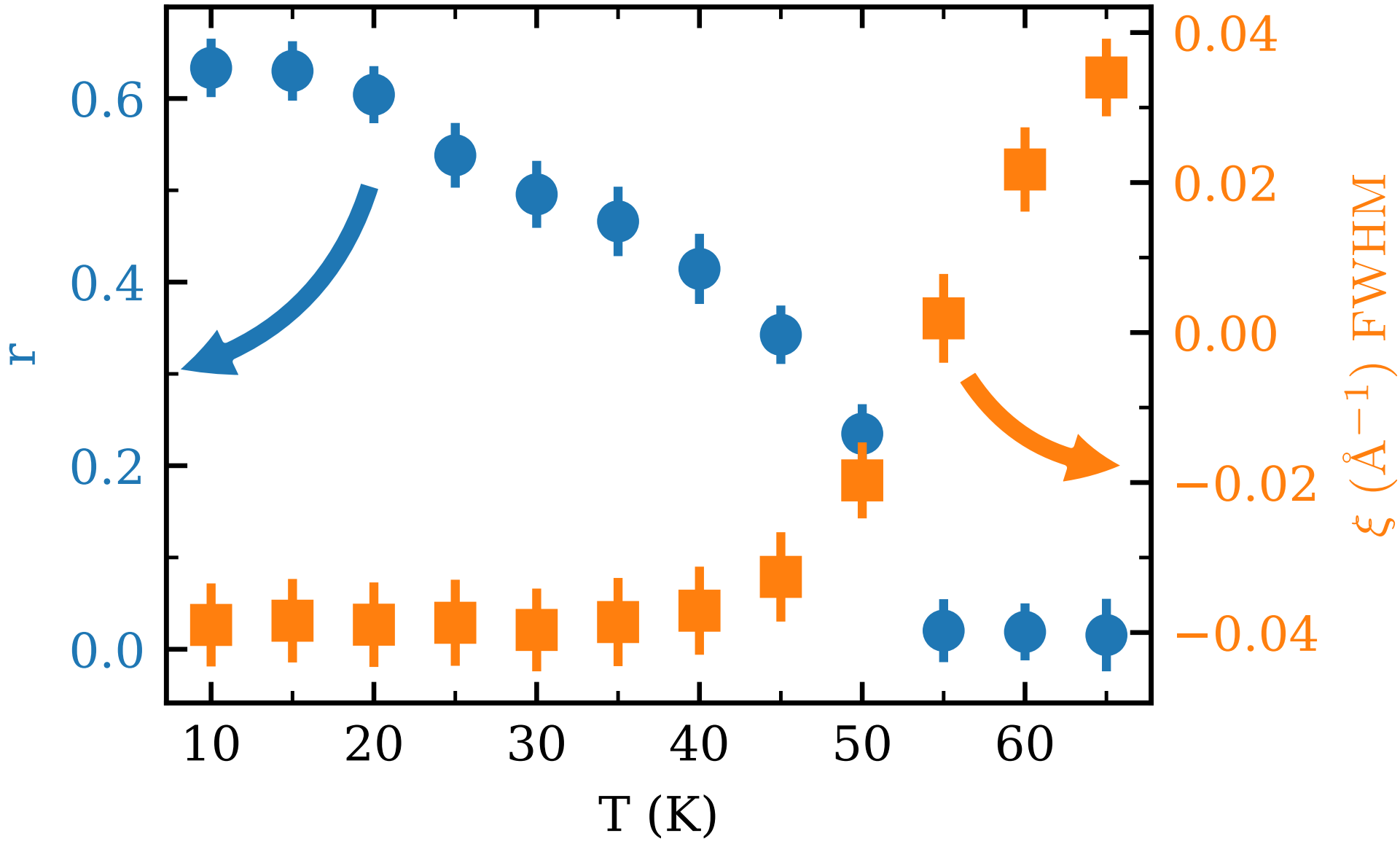
Isospin Up

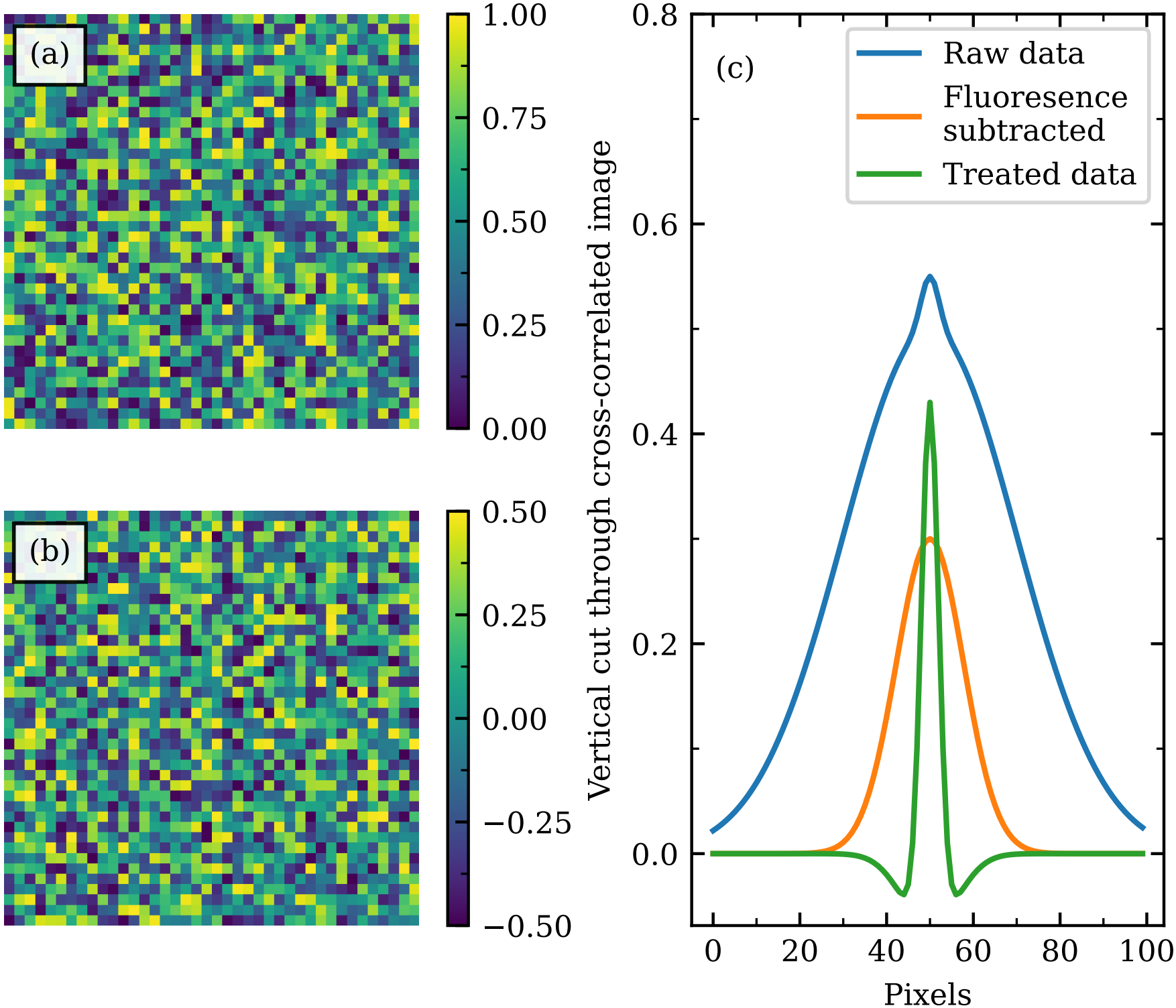


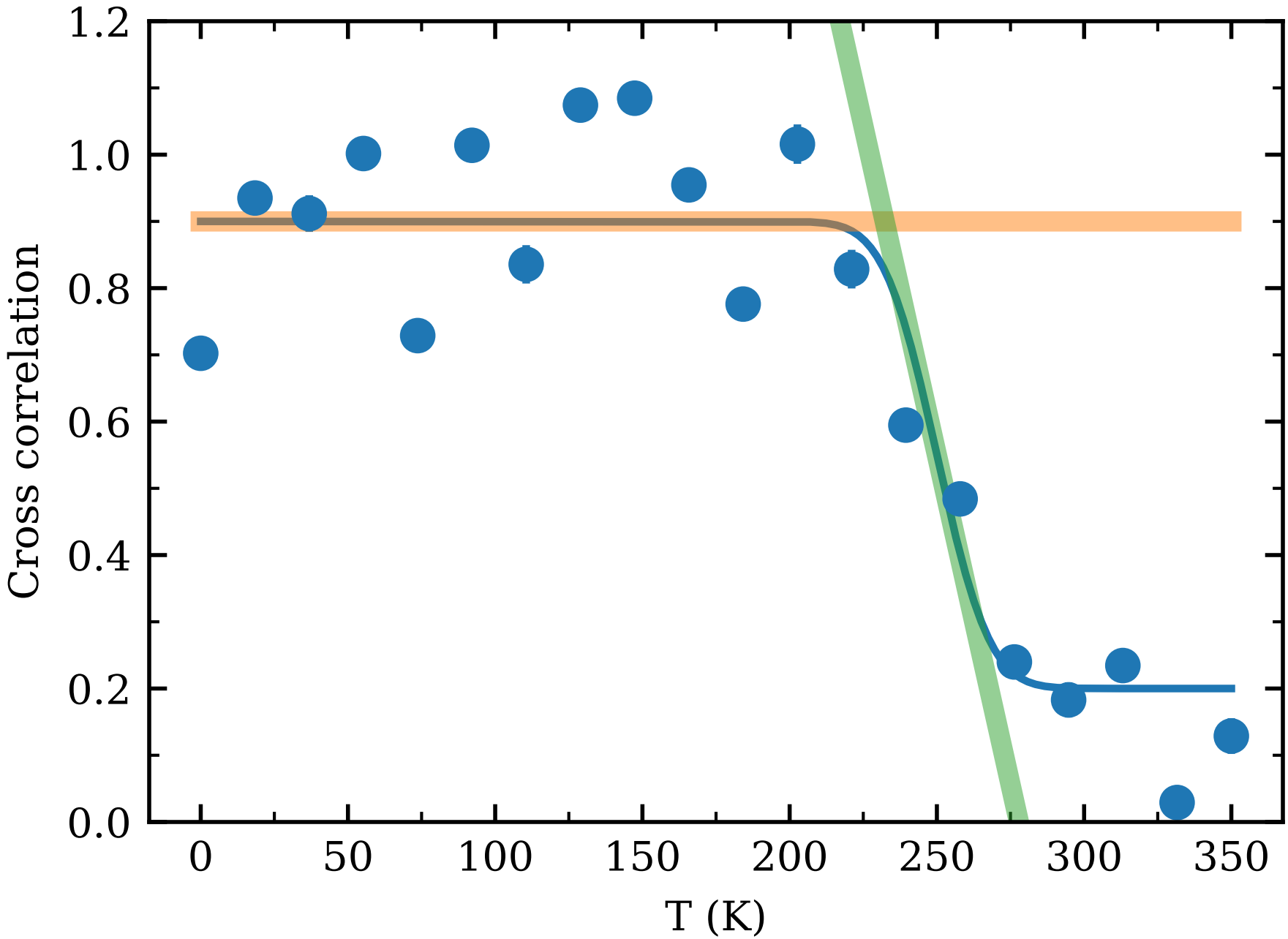
Isospin Down

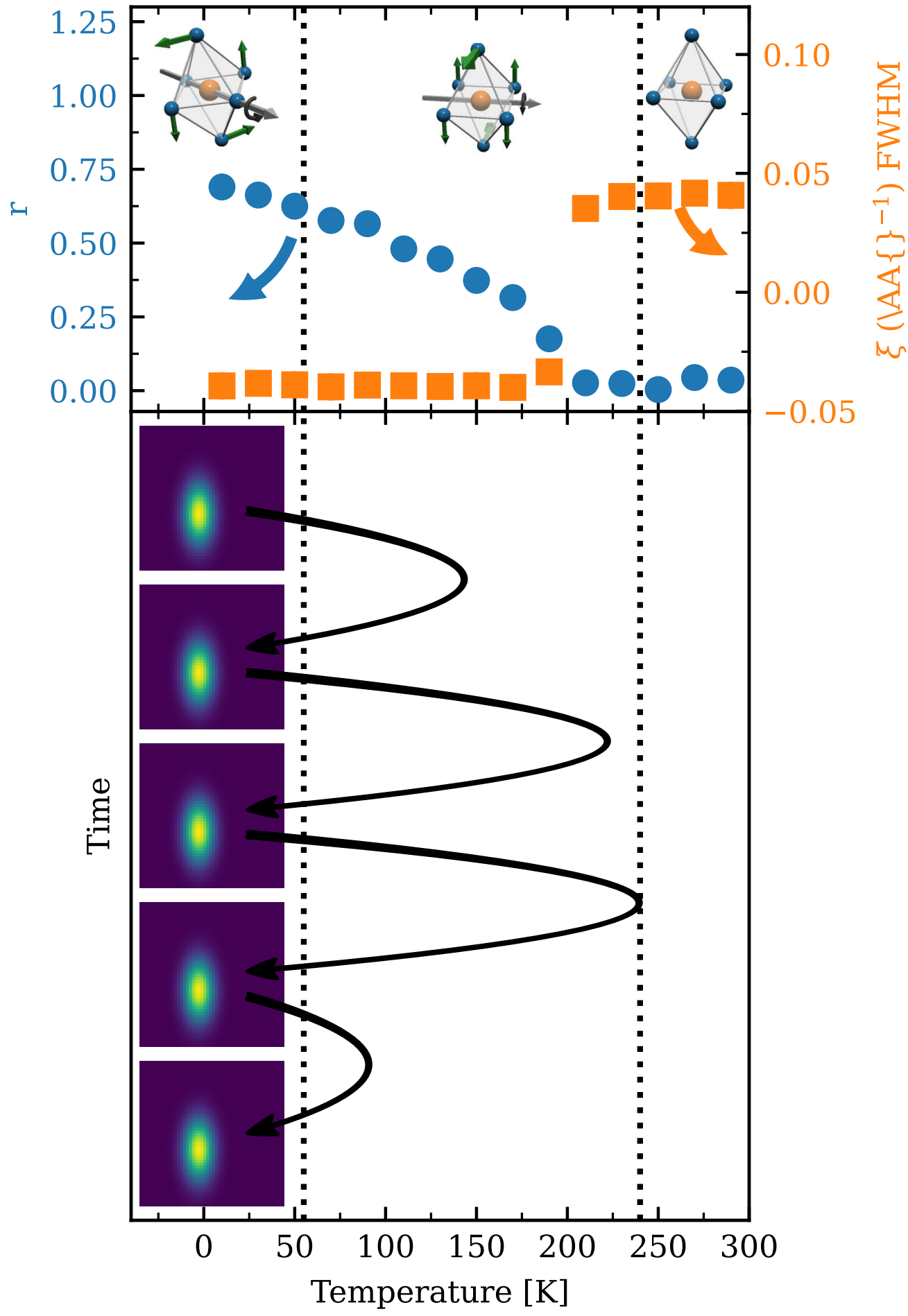


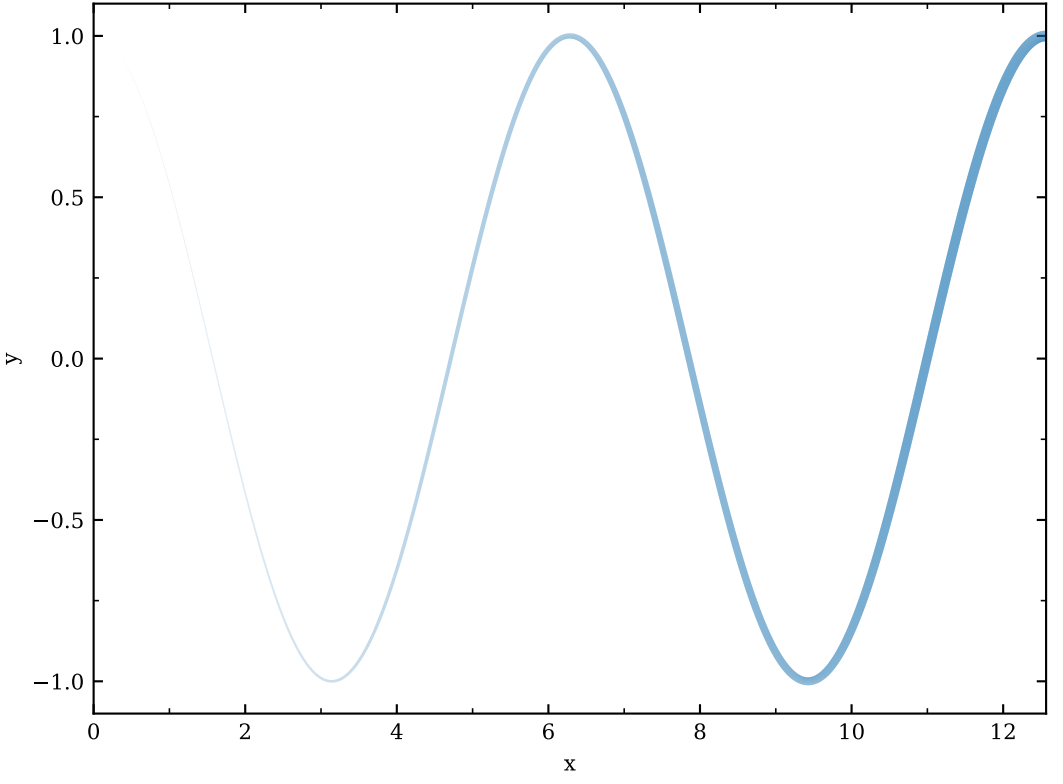

Domain size
$\sigma = 4 \pm 2$ pixels
$\sigma = 8 \pm 4$ pixels
$\sigma = 10 \pm 2$ pixels
CDW domains
Speckle pattern
(a)
(b)
(c)
(d)
(e)
(f)
Pixels
Pixels
0
5
10
15

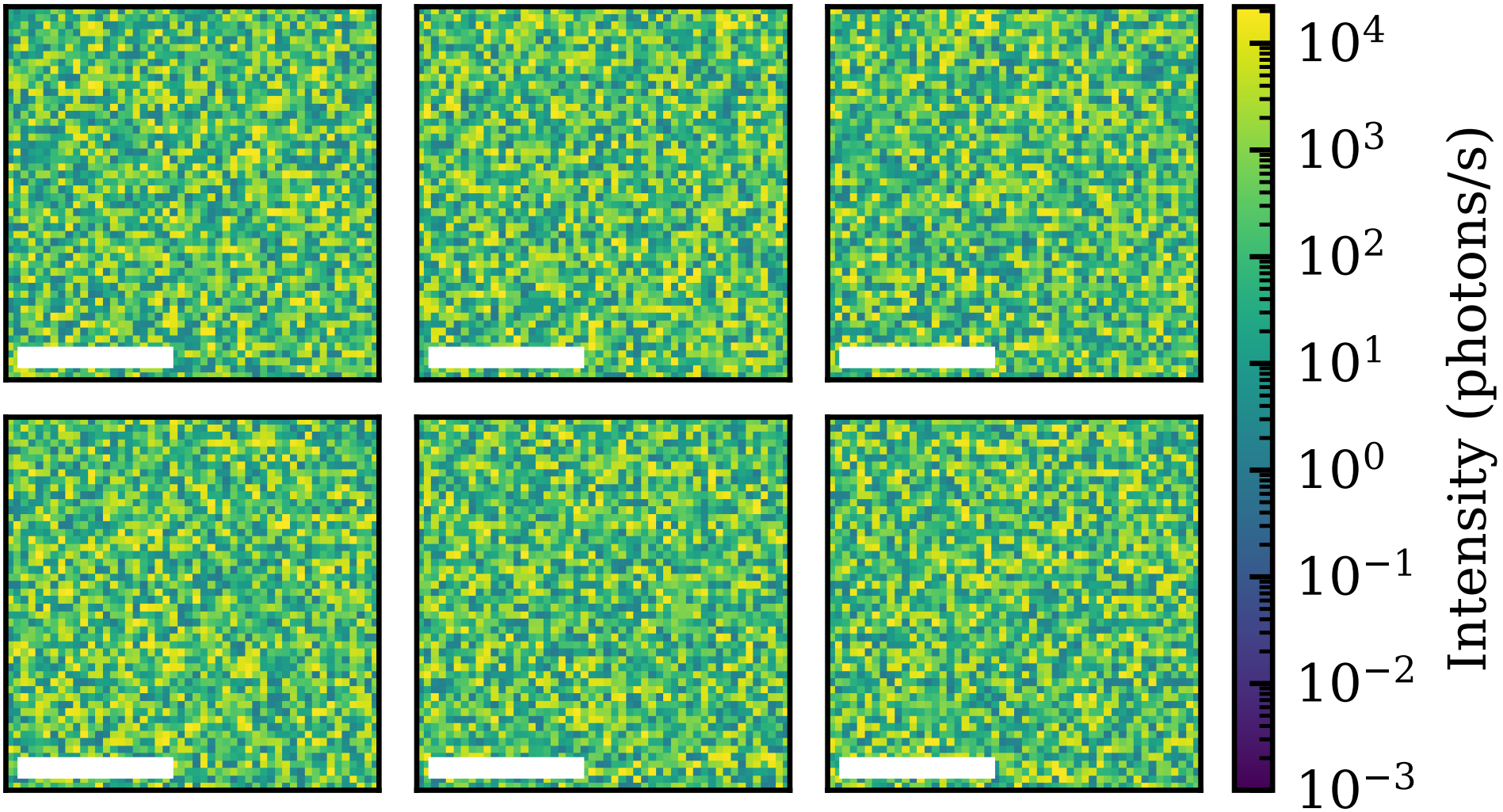

r
0.6
0.4
0.2
0.0
0.04
0.02
0.00
−0.02
−0.04
ξ (Å^{-1}) FWHM
10
20
30
40
50
60
T (K)

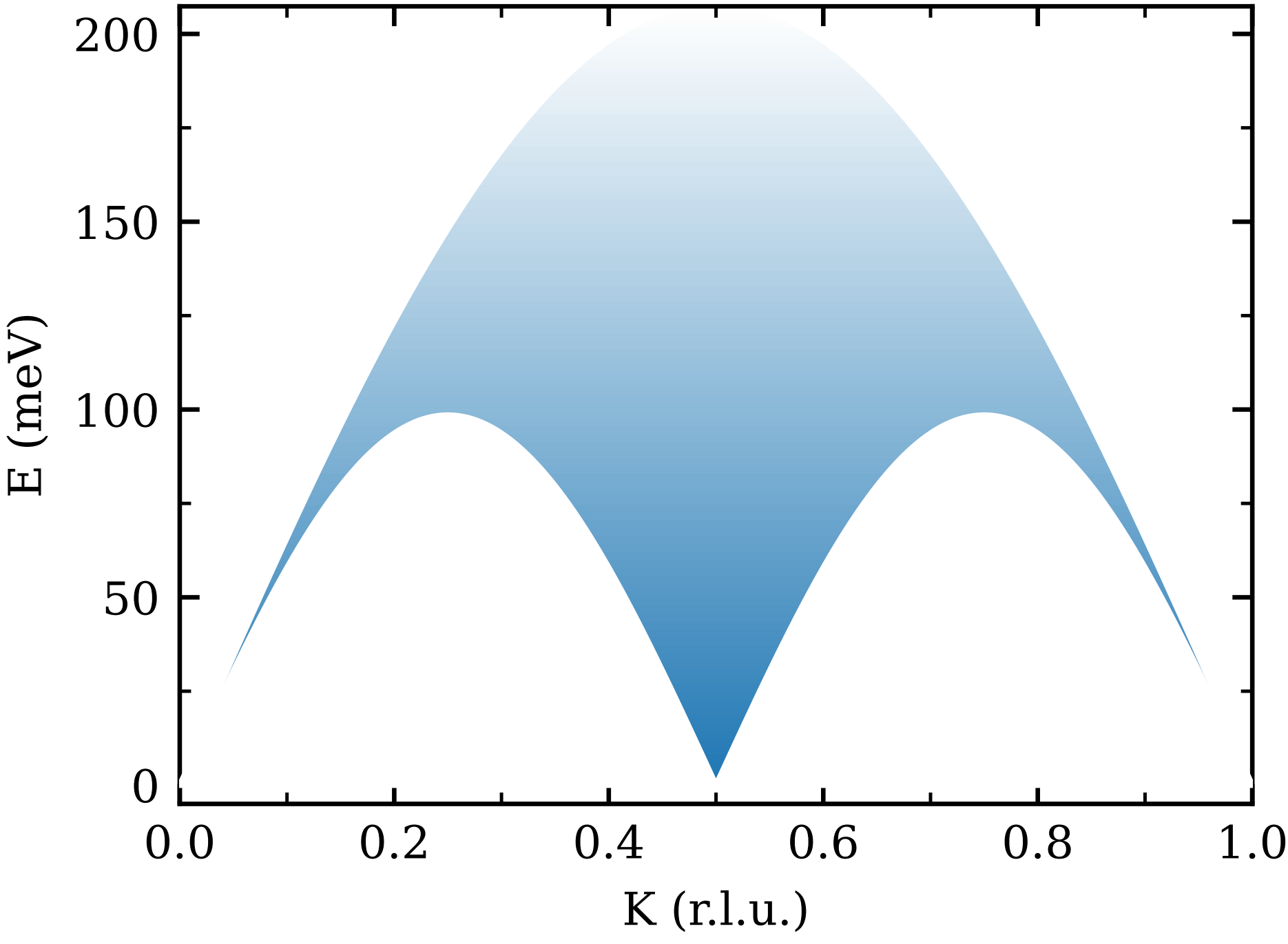

(a)
1.00
0.75
0.50
0.25
0.00
(b)
0.50
0.25
0.00
−0.25
−0.50
(c)
Raw data
Fluoresence subtracted
Treated data
Vertical cut through cross-correlated image
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Pixels

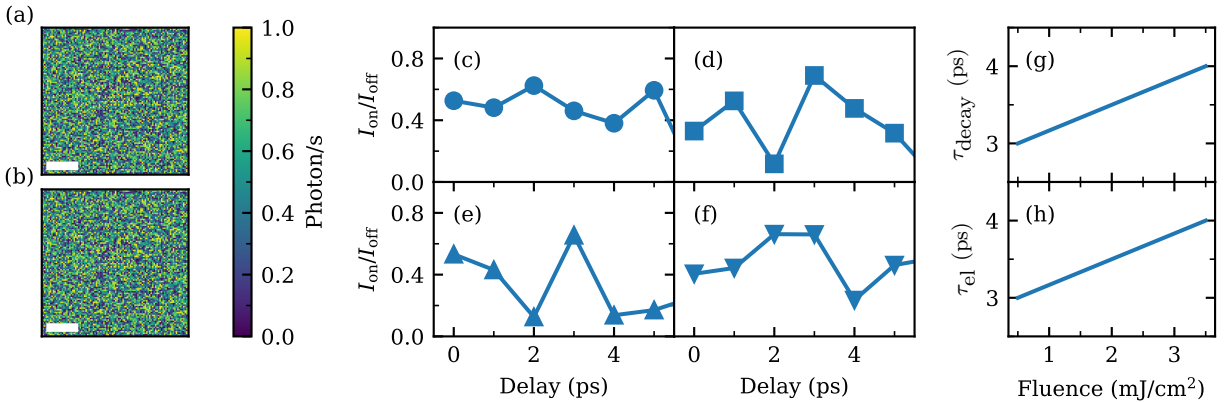

1.2
1.0
0.8
0.6
0.4
0.2
0.0
Cross correlation
0
50
100
150
200
250
300
350
T (K)

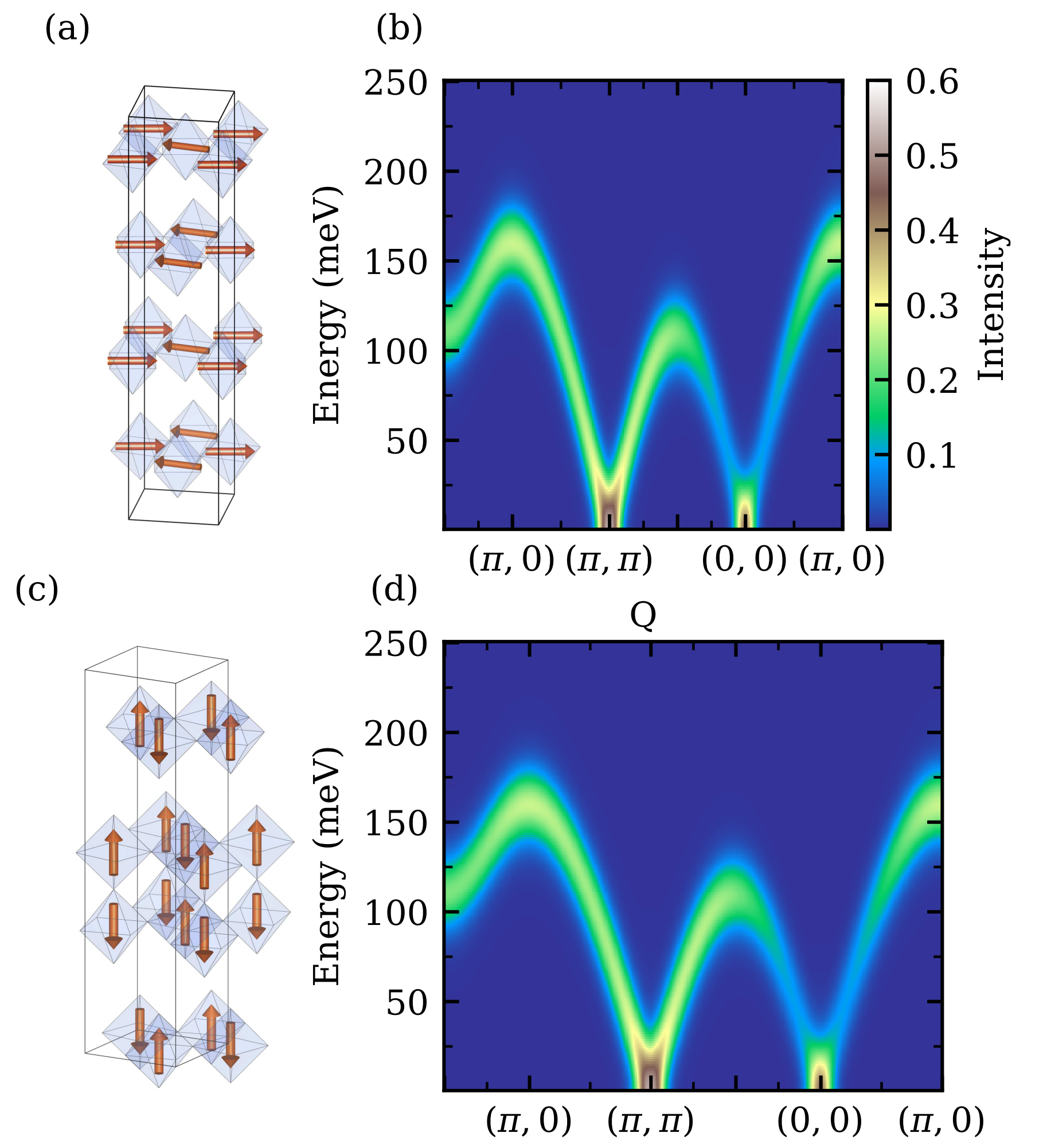

1.25
1.00
0.75
0.50
0.25
0.00
r
0.10
0.05
0.00
−0.05
ξ (\AA{}$^{-1}$) FWHM
Time
0
50
100
150
200
250
300
Temperature [K]

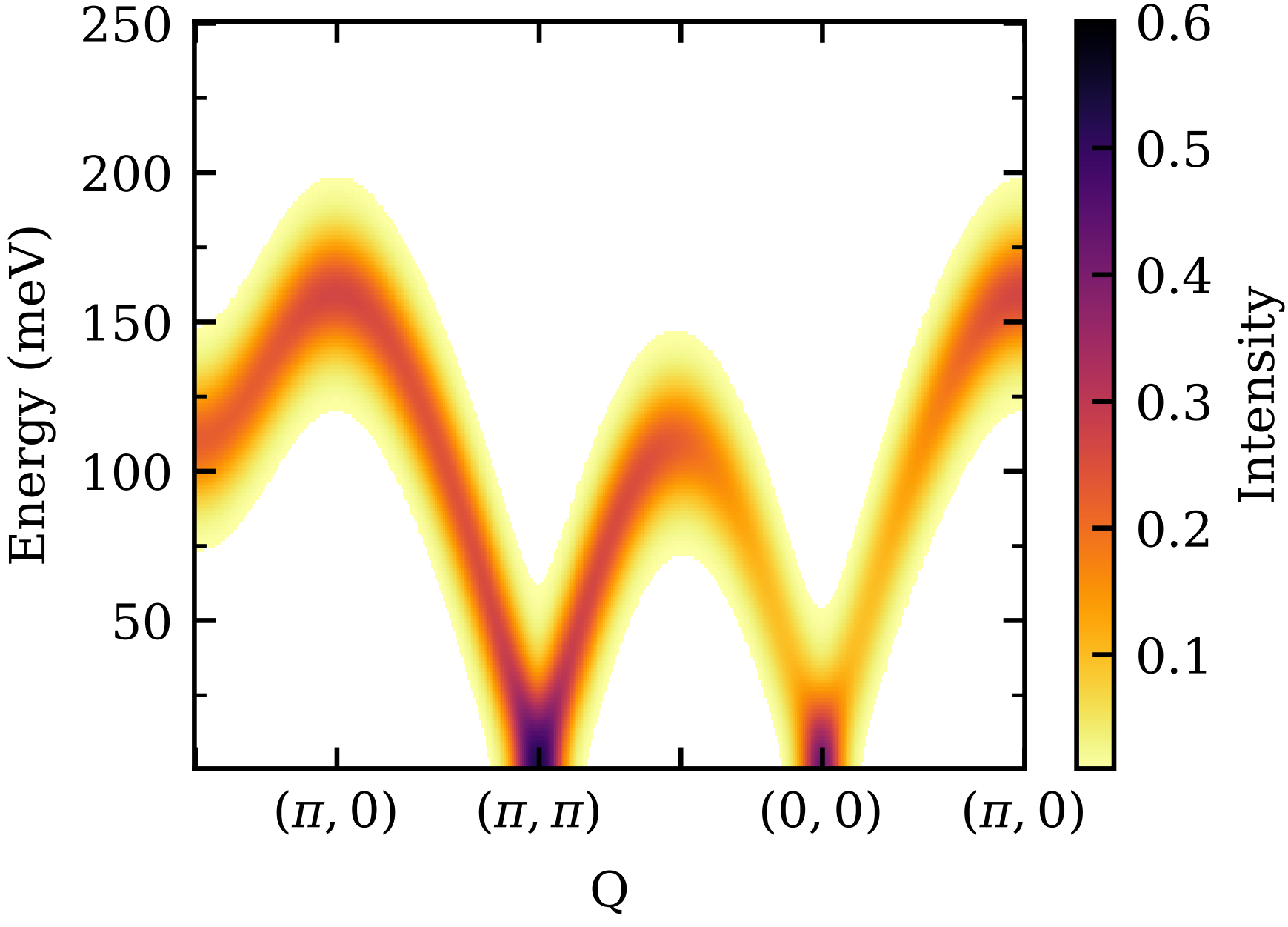

1.0
0.5
0.0
−0.5
−1.0
y
0
2
4
6
8
10
12
x

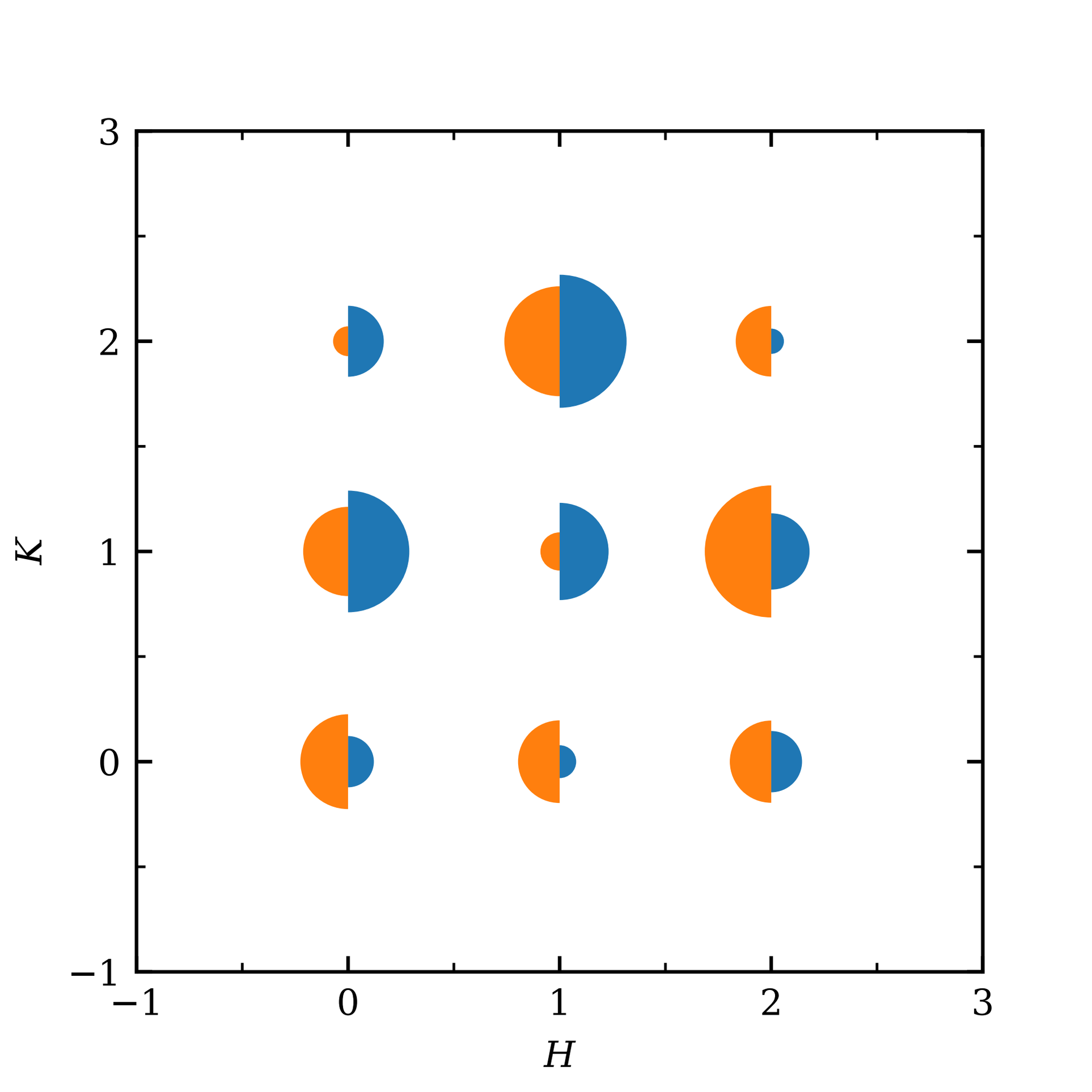

10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
Intensity (photons/s)

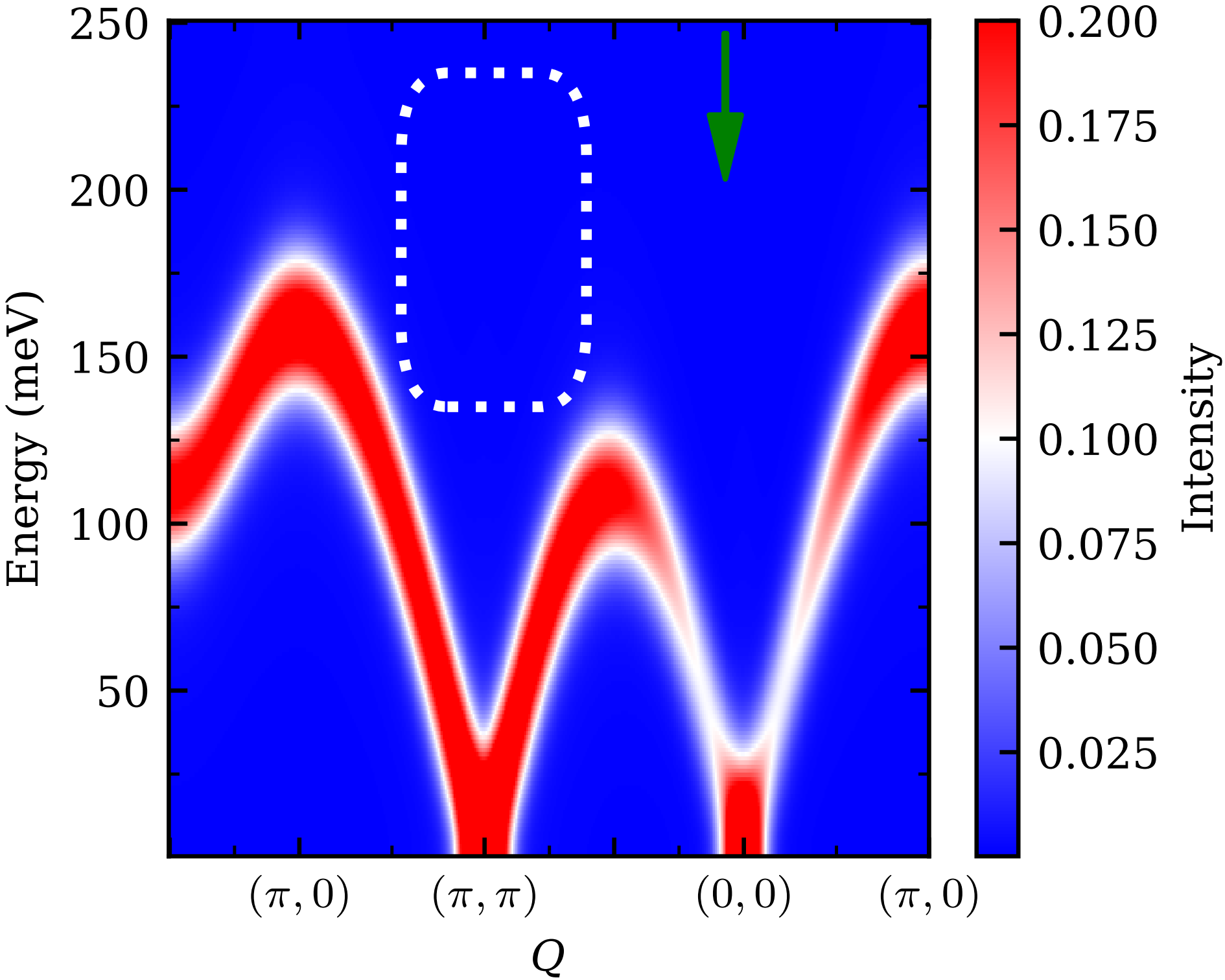

200
150
100
50
0
E (meV)
0.0
0.2
0.4
0.6
0.8
1.0
K (r.l.u.)

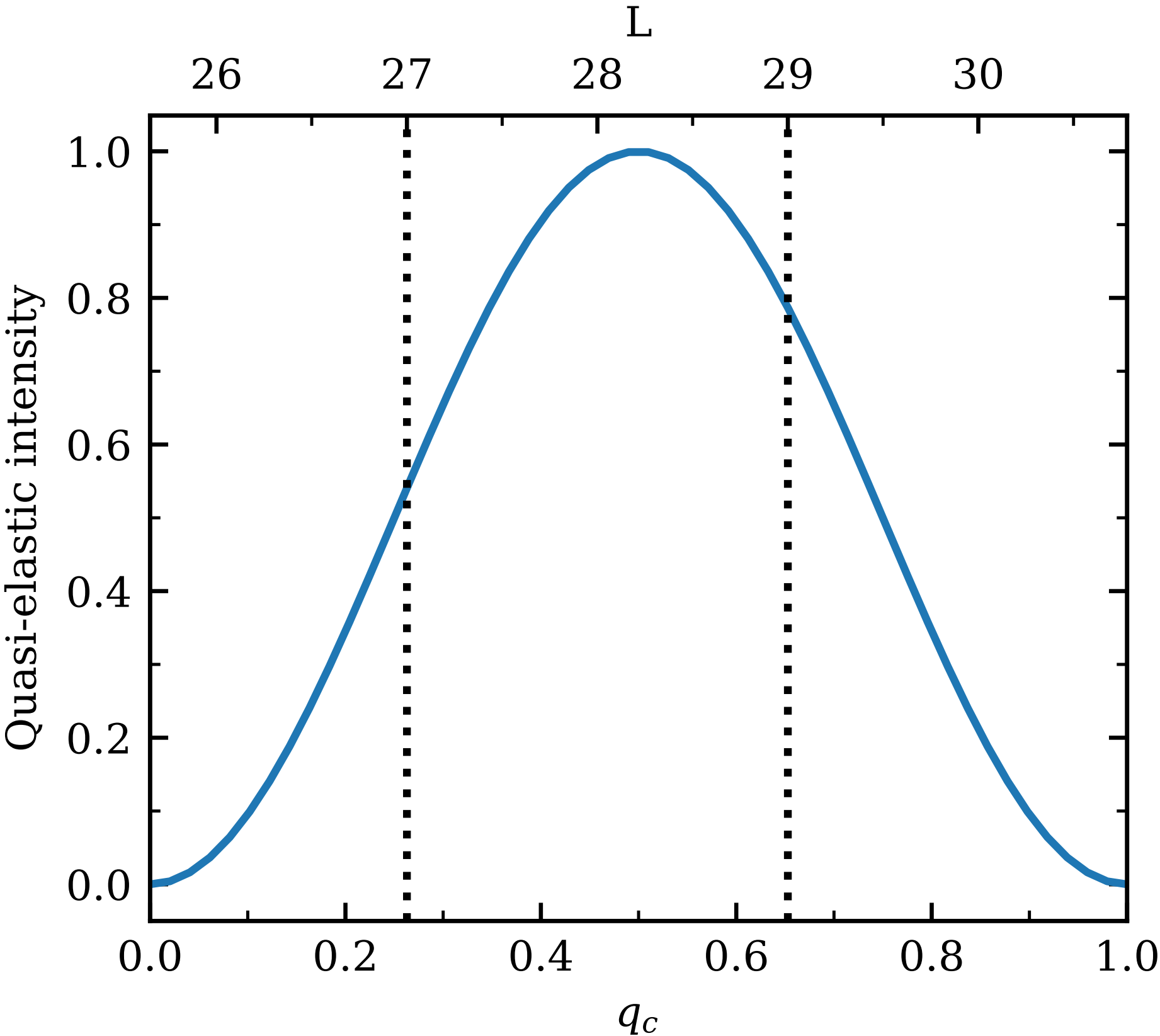

(a)
(b)
1.0
0.8
0.6
0.4
0.2
0.0
Photon/s
(c)
(d)
(e)
(f)
I_{on}/I_{off}
0.8
0.4
0.0
0
2
4
Delay (ps)
(g)
(h)
τ_{decay} (ps)
τ_{el} (ps)
4
3
1
2
3
Fluence (mJ/cm^2)

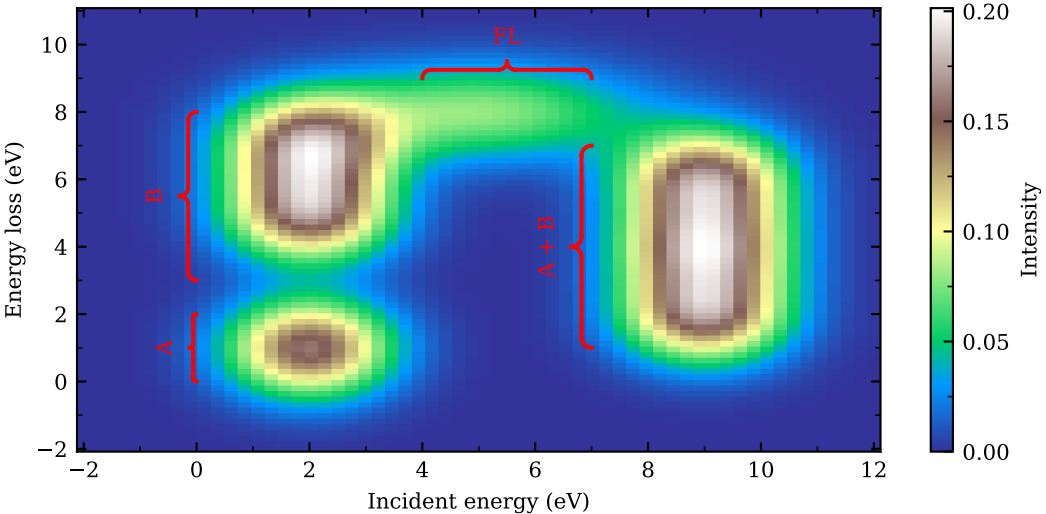

(a)
(b)
(c)
(d)
250
200
150
100
50
Energy (meV)
0.6
0.5
0.4
0.3
0.2
0.1
Intensity
(π, 0)
(π, π)
(0, 0)
(π, 0)
Q

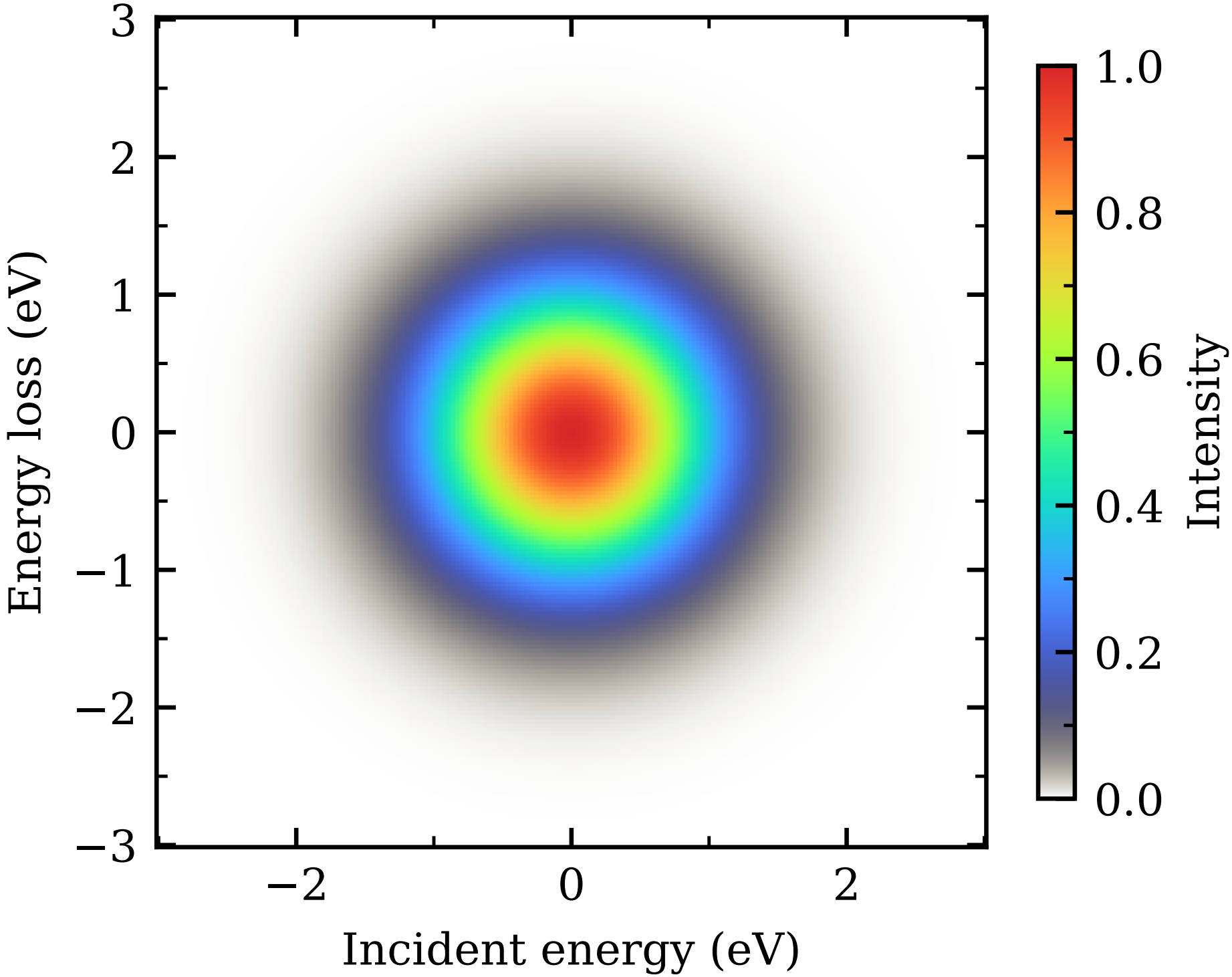

250
200
150
100
50
Energy (meV)
(π, 0)
(π, π)
(0, 0)
(π, 0)
Q
0.6
0.5
0.4
0.3
0.2
0.1
Intensity

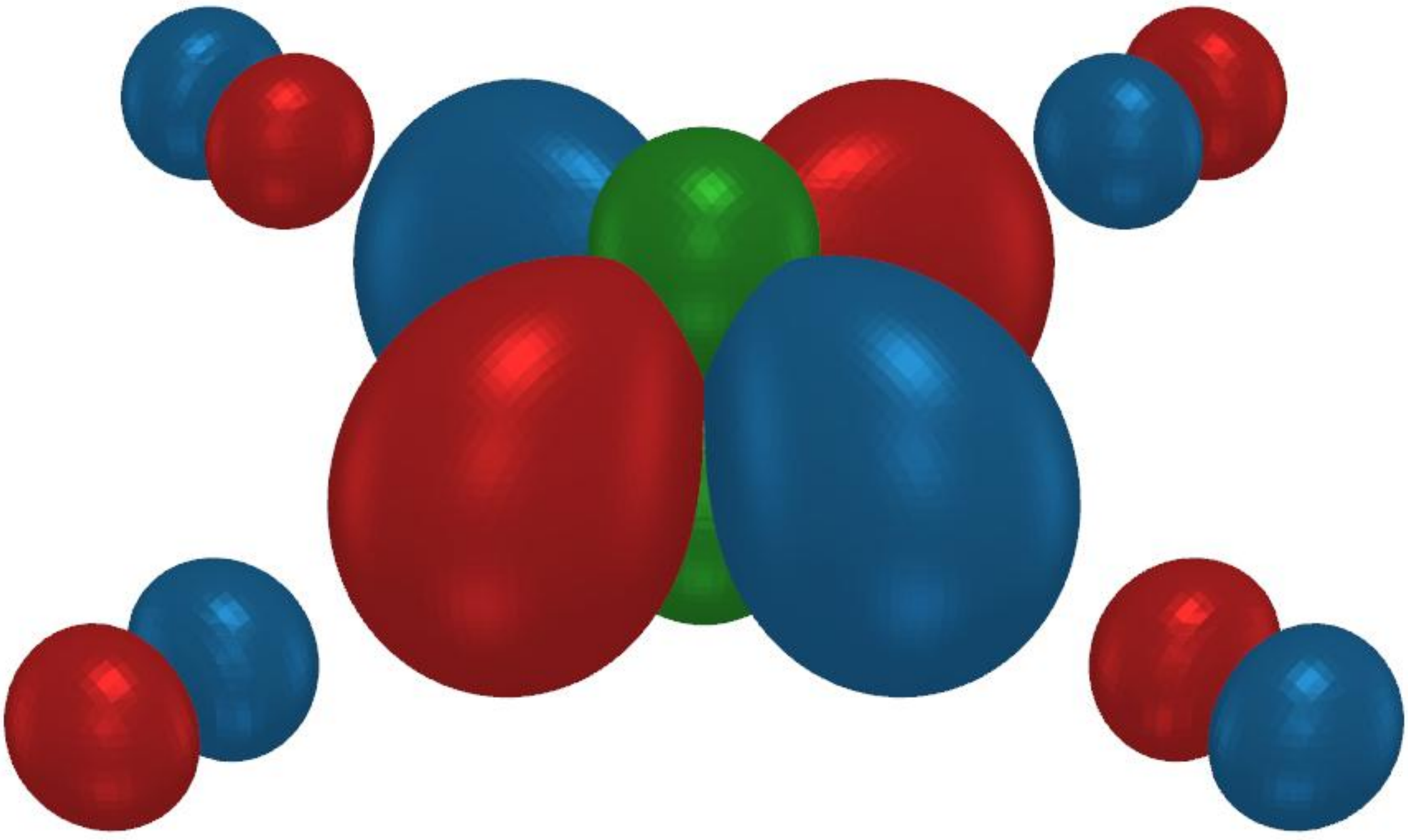

3
2
1
0
−1
−1
0
1
2
3
K
H

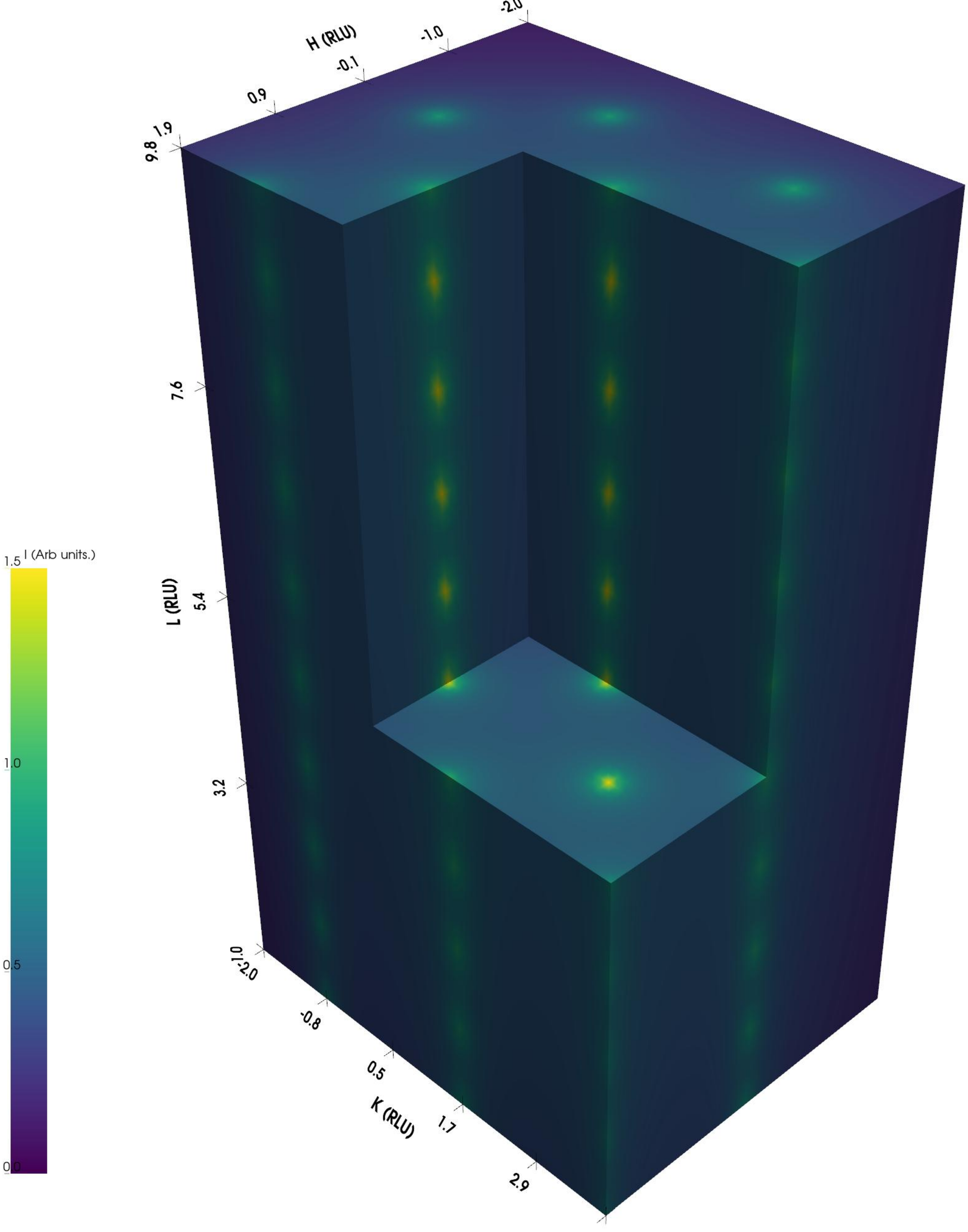

250
200
150
100
50
Energy (meV)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
Intensity
$(\pi, 0)$
(π, π)
$(0, 0)$
$(\pi, 0)$
Q

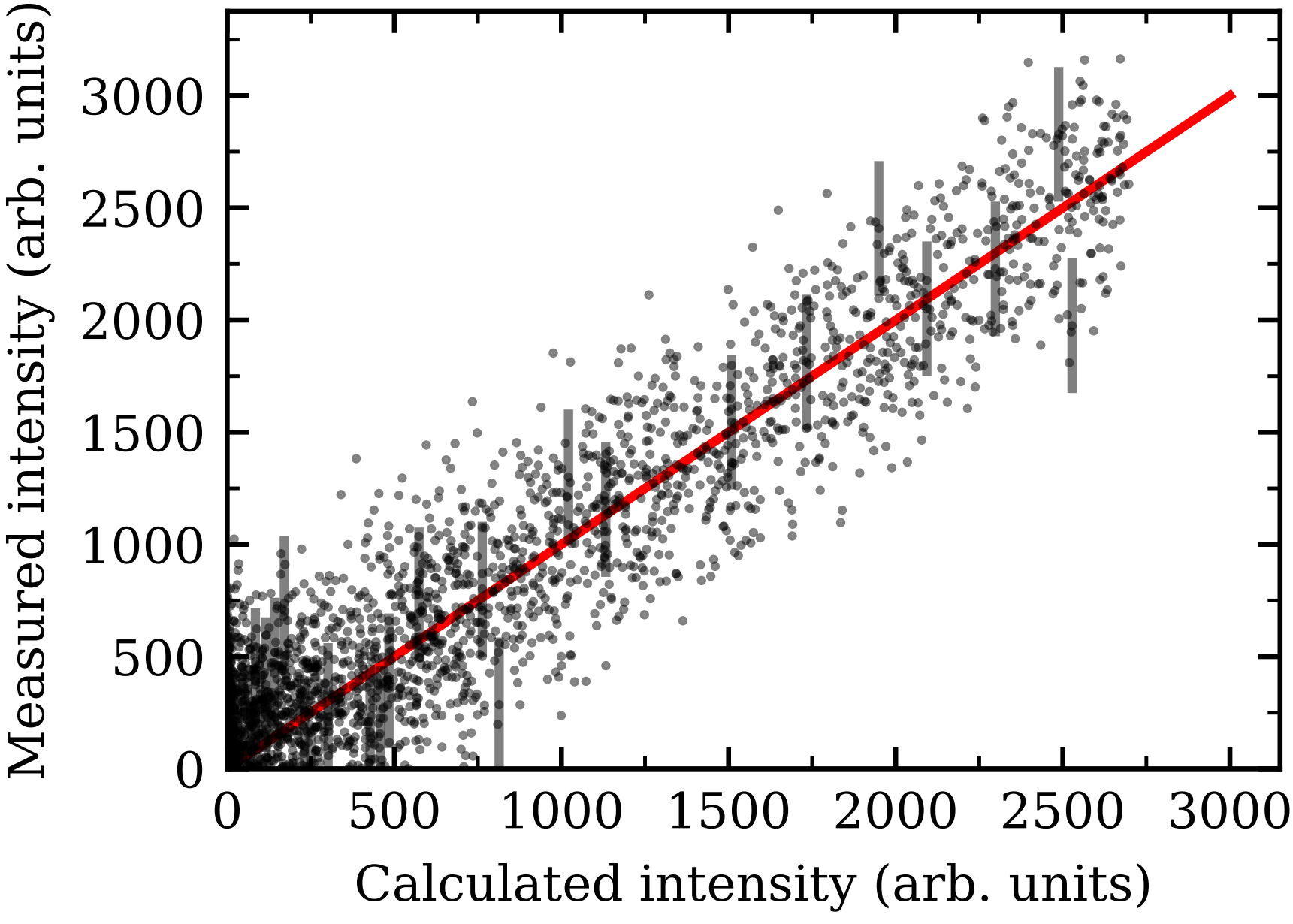

L
26
27
28
29
30
1.0
0.8
0.6
0.4
0.2
0.0
Quasi-elastic intensity
0.0
0.2
0.4
0.6
0.8
1.0
q_c

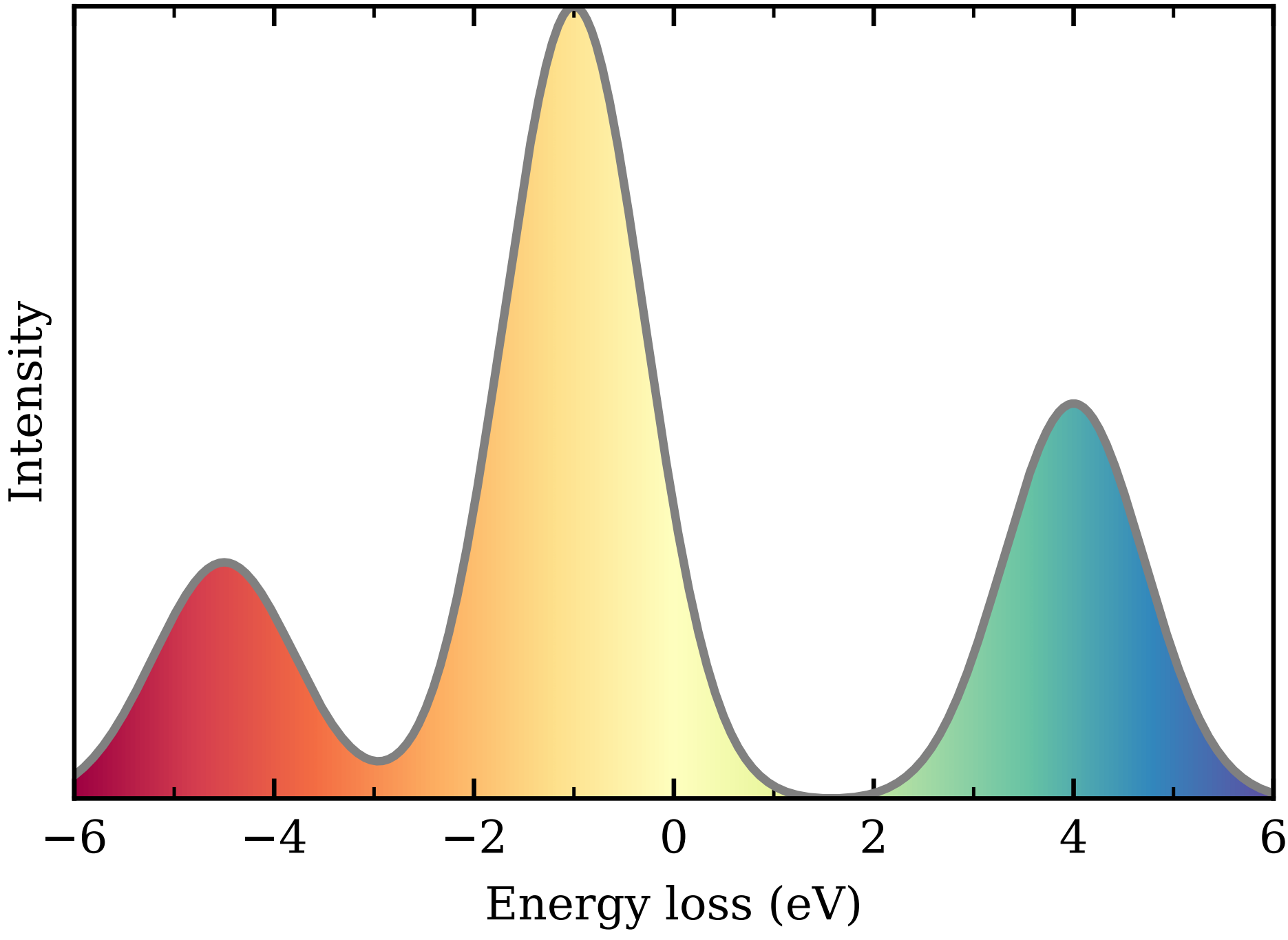

FL
B
A
A + B
Energy loss (eV)
Incident energy (eV)
Intensity
10
8
6
4
2
0
−2
−2
0
2
4
6
8
10
12
0.20
0.15
0.10
0.05
0.00

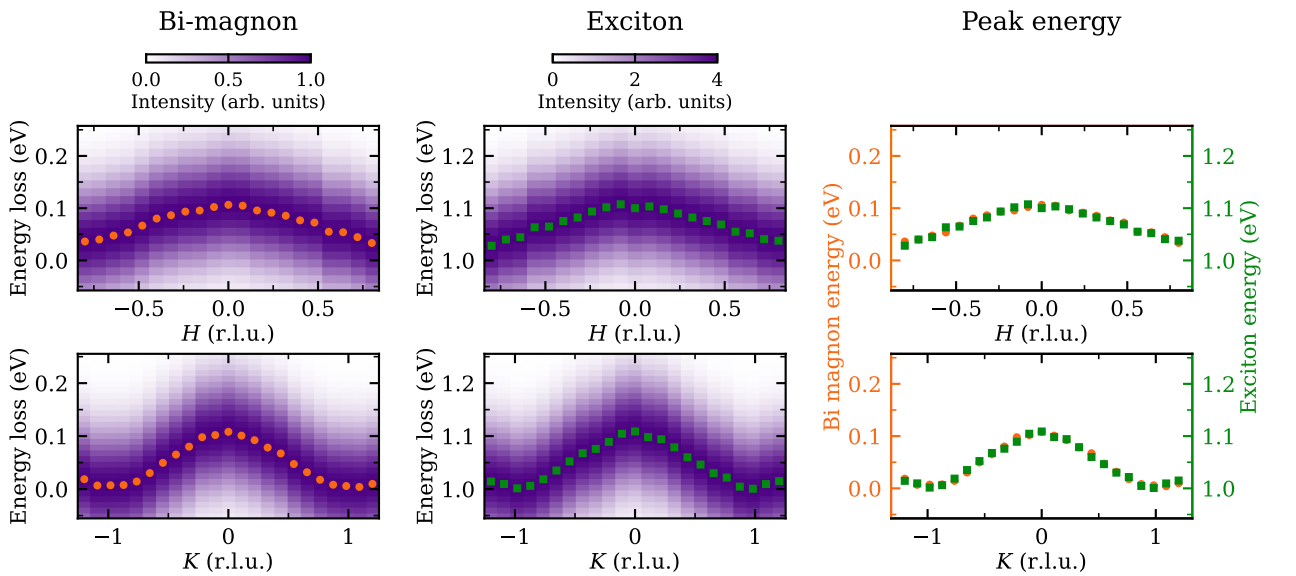

3
2
1
0
−1
−2
−3
Energy loss (eV)
−2
0
2
Incident energy (eV)
1.0
0.8
0.6
0.4
0.2
0.0
Intensity

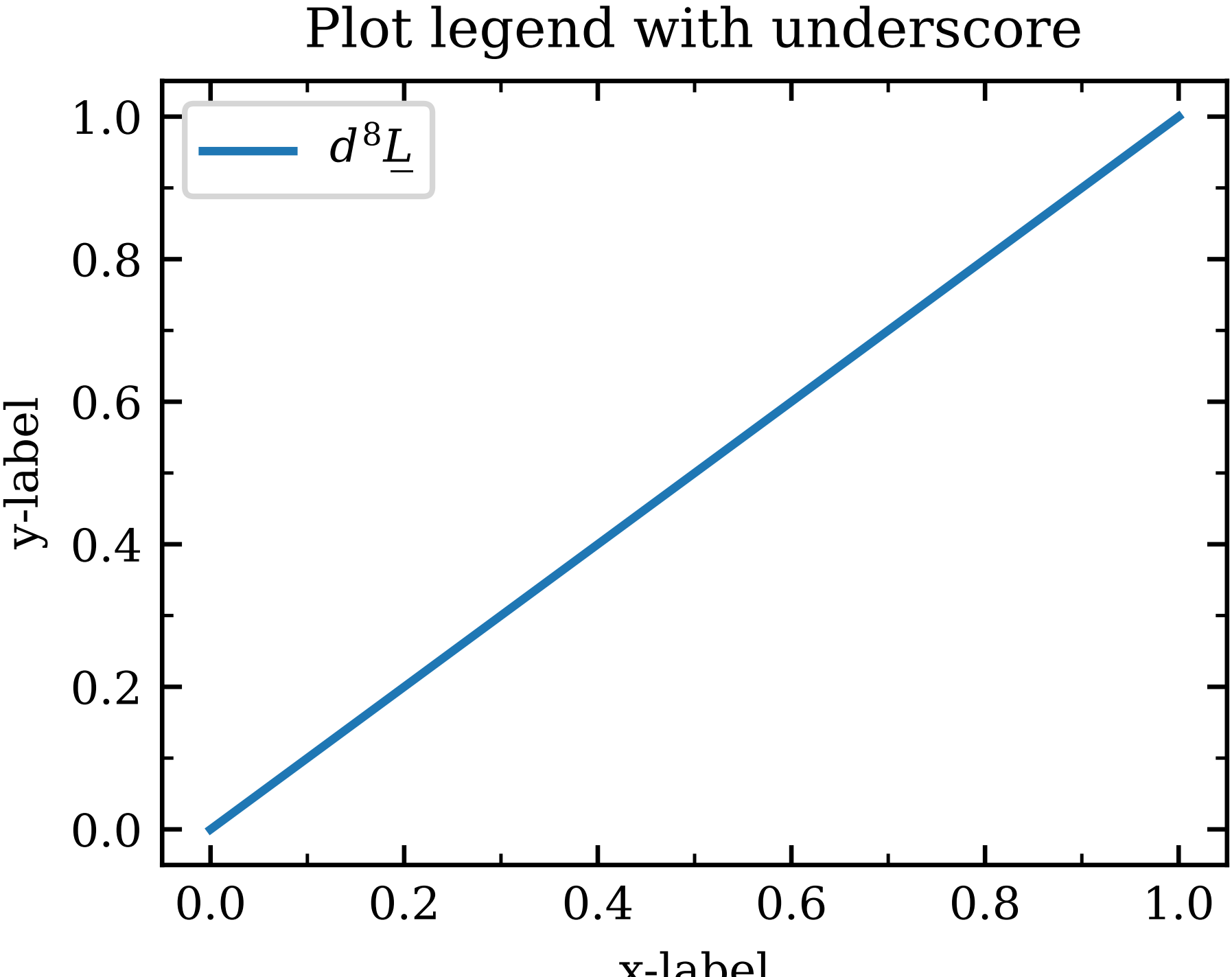


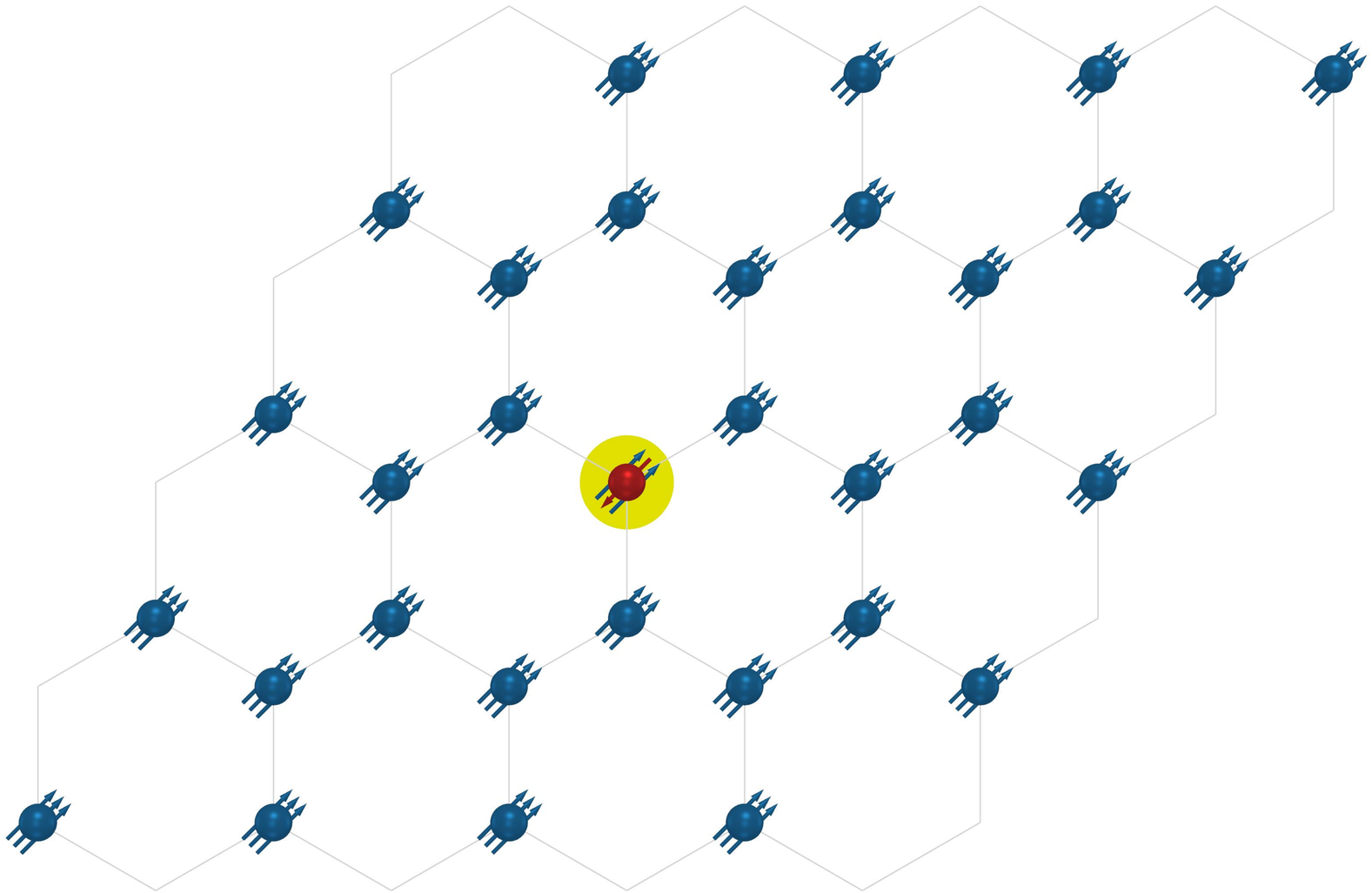

I (Arb units.)
1.5
1.0
0.5
0.0
H (RLU)
-2.0
-1.0
-0.1
0.9
1.9
9.8
L (RLU)
7.6
5.4
3.2
K (RLU)
-2.0
-1.0
-0.8
0.5
1.7
2.9

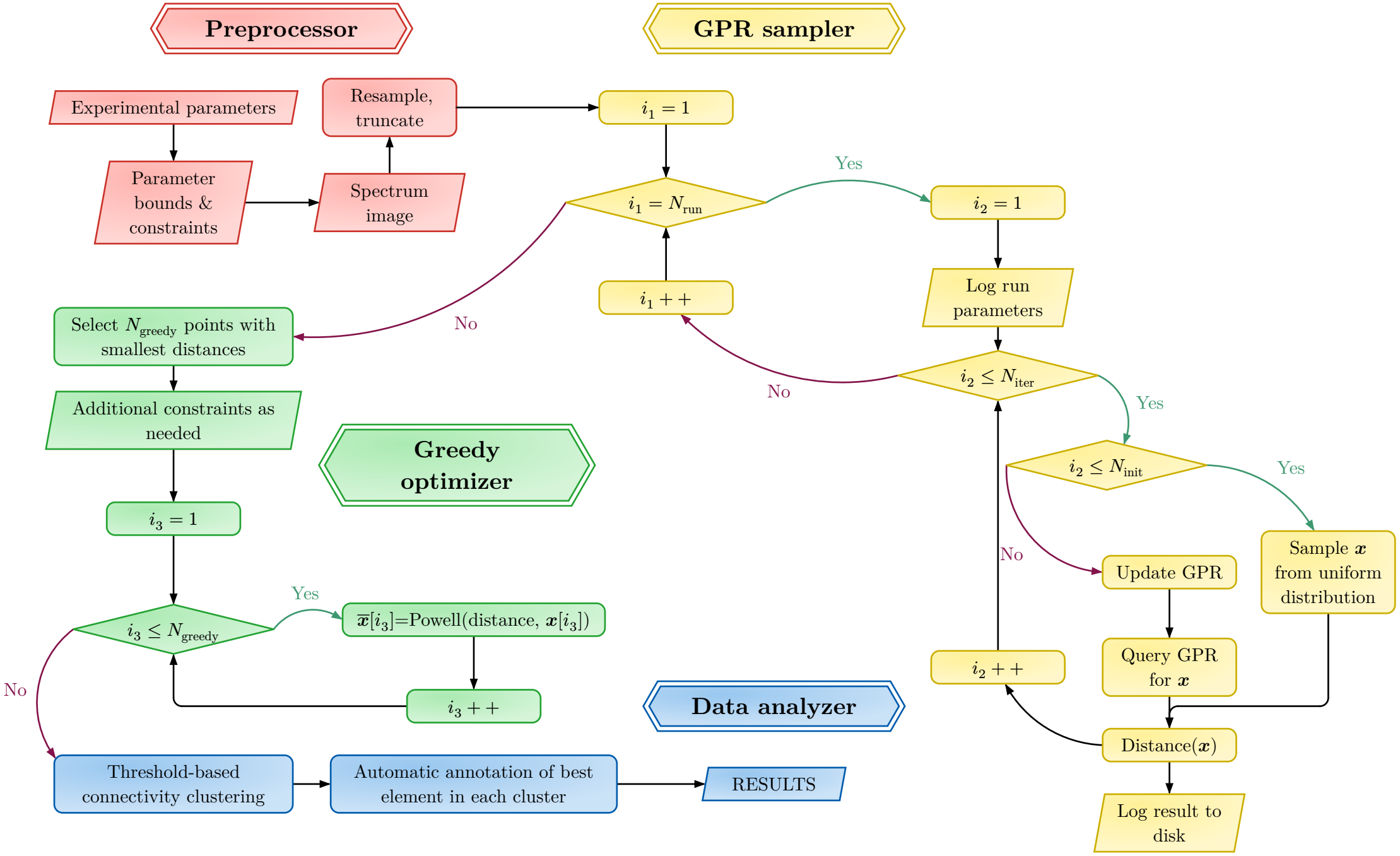

Measured intensity (arb. units)
0
500
1000
1500
2000
2500
3000
Calculated intensity (arb. units)
0
500
1000
1500
2000
2500
3000

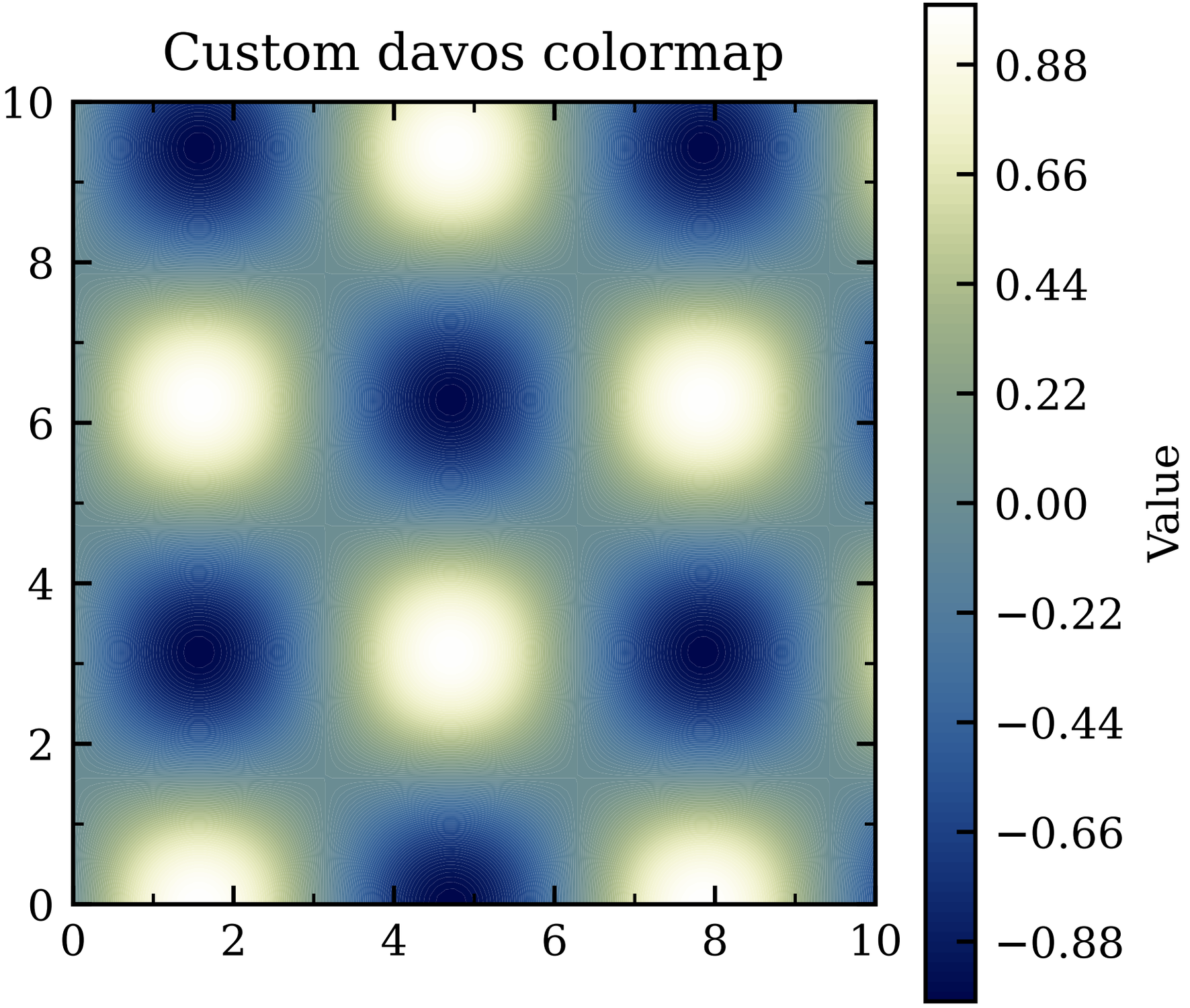

Intensity
−6
−4
−2
0
2
4
6
Energy loss (eV)

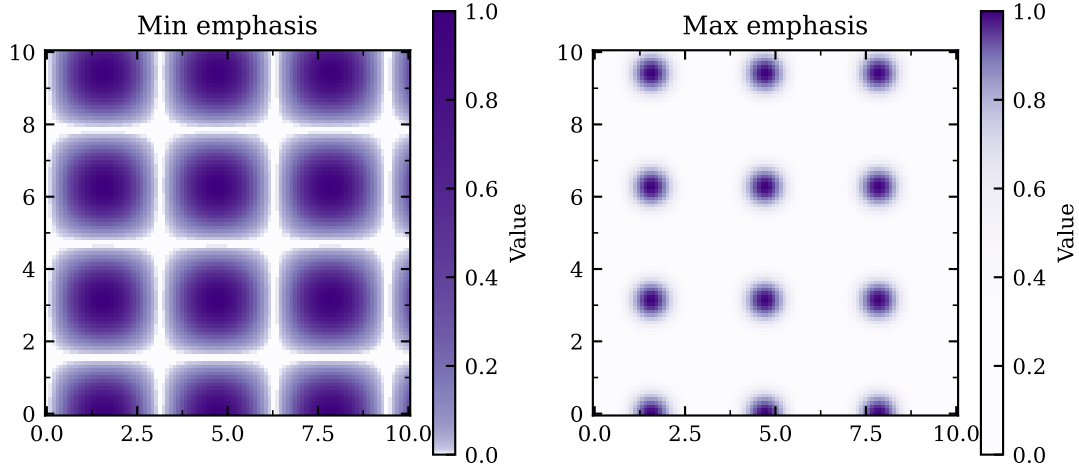

Bi-magnon
Exciton
Peak energy
0.0
0.5
1.0
Intensity (arb. units)
0
2
4
Intensity (arb. units)
Energy loss (eV)
0.2
0.1
0.0
1.2
1.1
1.0
−0.5
0.0
0.5
H (r.l.u.)
−1
0
1
K (r.l.u.)
Bi magnon energy (eV)
Exciton energy (eV)

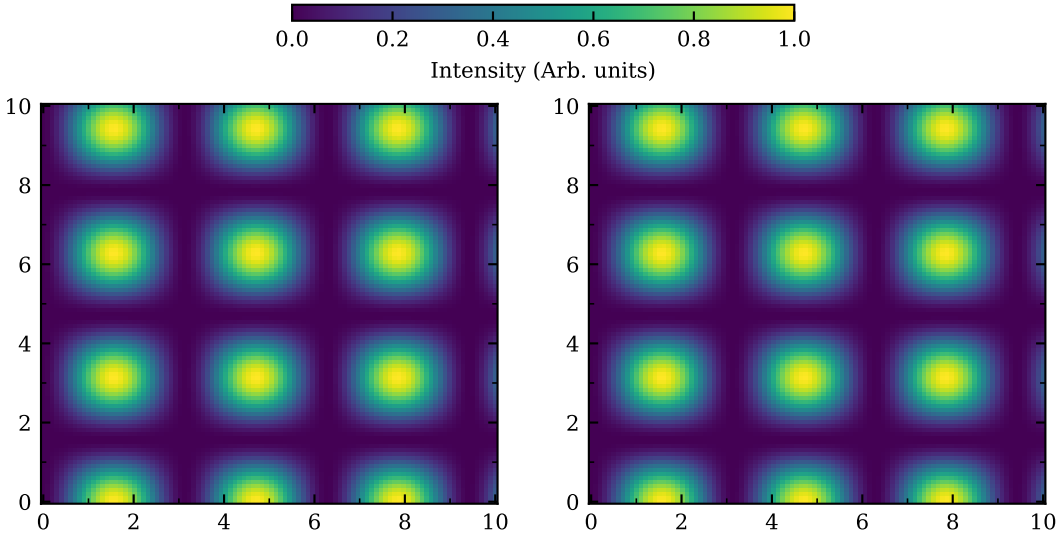

Plot legend with underscore
$d^8\underline{L}$
1.0
0.8
0.6
0.4
0.2
0.0
y-label
0.0
0.2
0.4
0.6
0.8
1.0
x-label




Preprocessor
GPR sampler
Experimental parameters
Parameter bounds & constraints
Spectrum image
Resample, truncate
$i_1 = 1$
$i_1 = N_{\text{run}}$
Yes
No
$i_1 + +$
$i_2 = 1$
Log run parameters
$i_2 \leq N_{\text{iter}}$
No
Yes
$i_2 \leq N_{\text{init}}$
Yes
No
Update GPR
Query GPR for $\boldsymbol{x}$
Distance($\boldsymbol{x}$)
Log result to disk
Sample $\boldsymbol{x}$ from uniform distribution
$i_2 + +$
Select N_{greedy} points with smallest distances
Additional constraints as needed
Greedy optimizer
$i_3 = 1$
$i_3 \leq N_{\text{greedy}}$
Yes
No
$\overline{\boldsymbol{x}}[i_3]$=Powell(distance, $\boldsymbol{x}[i_3]$)
$i_3 + +$
Data analyzer
Threshold-based connectivity clustering
Automatic annotation of best element in each cluster
RESULTS


Custom davos colormap
10
8
6
4
2
0
0
2
4
6
8
10
0.88
0.66
0.44
0.22
0.00
−0.22
−0.44
−0.66
−0.88
Value


Min emphasis
Max emphasis
Value
Value
0.0
2.5
5.0
7.5
10.0
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0


0.0
0.2
0.4
0.6
0.8
1.0
Intensity (Arb. units)
0
2
4
6
8
10